

When Can You Trust Linear Regression?

An Investigation of the Robustness of Ordinary Least Squares Under Real-World Data Conditions

Abstract

Ordinary Least Squares (OLS) is simple to fit and easy to misuse: its formulas return an answer regardless of whether the data satisfy the conditions that make that answer meaningful. This report treats OLS as a scientific instrument and asks, empirically, when it can be trusted. Four controlled Monte Carlo experiments — each varying one property of the data while holding a common baseline model fixed — quantify how OLS's coefficients, confidence intervals, and predictive accuracy respond to shrinking sample size, response outliers, predictor multicollinearity, and unmodeled curvature. Three of these four violations leave the average coefficient estimate essentially unbiased and instead damage precision or prediction; the exception is nonlinearity when the omitted curvature correlates with an included predictor. Two of the experiments are then extended head-to-head against a standard robust alternative — Huber regression against outliers, Ridge regression against multicollinearity — to test whether switching methods helps as much as the diagnostics suggest it should. Finally, the resulting checklist is applied to a real dataset (the scikit-learn diabetes cohort) with unknown ground truth, to test whether the simulation-derived intuitions transfer. They do: elevated Variance Inflation Factors flag exactly the collinear predictor block, and residual-based tests (Breusch–Pagan, Ramsey RESET) catch mild heteroscedasticity and nonlinearity that the coefficient table alone does not reveal.

Introduction

OLS is the most-used statistical procedure in applied work, and under the Gauss–Markov conditions it is probably the best linear unbiased estimator. But software functions like `lm()` and `LinearRegression().fit()` return coefficients, standard errors, and p-values whether or not those conditions hold — the procedure never warns you when it shouldn't be trusted.

Assumption violations are also the rule rather than the exception in real data: small samples, contaminated measurements, correlated predictors, and genuinely curved relationships are all commonplace, and they do not all cause the same kind of damage. Treating “check your assumptions” as a single pass/fail gate risks over-reacting to a harmless issue or ignoring a serious one — the useful question is which aspect of trust (a point estimate, its stated uncertainty, or its predictions) each violation actually threatens.

Real data cannot answer that question directly, because the true coefficients and functional form are unknown — there is no ground truth to compare a fitted line against. Simulation supplies that ground truth: by generating data from a model we fully control, we can measure exactly how far OLS's estimates drift as we deliberately break one assumption at a time. This report does that, then checks whether the resulting intuitions actually matter in practice — first by testing a robust alternative head-to-head against OLS for two of the violations, and second by applying the resulting checklist to a real dataset.

Research Question

Central question:

“Under what conditions should a researcher stop trusting the conclusions produced by Ordinary Least Squares Linear Regression?”

This is decomposed into six sub-questions, one per experiment:

- 1. Sample size** — How much data is needed before OLS produces stable, trustworthy estimates?
- 2. Outliers** — How sensitive is OLS to an increasing share of contaminated observations?
- 3. Multicollinearity** — How does increasing correlation between predictors affect coefficient reliability?
- 4. Nonlinearity** — What happens when a linear model is fit to a relationship that is not actually linear?
- 5. Method choice** — Once a violation is diagnosed, does switching to a robust estimator (Huber, Ridge) actually help, and by how much?
- 6. External validity** — Do these simulation-derived intuitions transfer to a real dataset with unknown ground truth?

Every experiment, figure, and interpretation below is organized around answering one of these six questions, in service of the central question above.

Background on Linear Regression

The linear model relates a response y to predictors X through $y = X\beta + \varepsilon$. Ordinary Least Squares chooses the coefficient vector β that minimizes the sum of squared residuals, which has the closed-form solution $\beta = (X'X)^{-1}X'y$.

Under the Gauss–Markov conditions — linearity in parameters, independent errors with constant variance (homoscedasticity), and no perfect collinearity among predictors — β is the Best Linear Unbiased Estimator (BLUE): unbiased, and lower-variance than any other linear unbiased estimator. If the errors are additionally assumed Normal, β is also the maximum-likelihood estimator, and exact finite-sample t - and F -tests are available; without Normality, inference is still justified for large n via the Central Limit Theorem.

Every guarantee in the preceding paragraph is conditional. This report investigates what happens to those guarantees, quantitatively, as their preconditions are relaxed one at a time.

Assumptions of OLS

#	Assumption	In plain terms
1	Linearity	The conditional mean of y is a linear function of X
2	Exogeneity	$E[\varepsilon X] = 0$ — errors are uncorrelated with predictors
3	No perfect multicollinearity	No predictor is an exact linear combination of the others
4	Homoscedasticity	Error variance is constant across all values of X
5	Independence	Errors are uncorrelated across observations

#	Assumption	In plain terms
6	Normality (for exact inference)	Errors are Normally distributed (or n is large enough for the CLT to apply)
7	Adequate sample size	Enough observations relative to the number of parameters being estimated

This report focuses on four of these — sample size (7), outliers (which violate Normality/homoscedasticity in practice, 4 & 6), multicollinearity (3), and nonlinearity (1) — because they are the violations a working statistician encounters most often and, in our experience, most often under-diagnoses. Independence violations (e.g., time-series autocorrelation) are noted as a natural extension beyond this report's scope.

Simulation Framework

Every experiment shares one baseline data-generating process (DGP), representing the ideal conditions under which OLS's guarantees hold: $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$, $\varepsilon \sim N(0, \sigma^2)$, with $\beta_0 = 5$, $\beta_1 = 2$, $\beta_2 = 1.5$, $\sigma = 2.5$, and $X_1, X_2 \sim N(0,1)$ independently. β_1 — the coefficient on X_1 — is tracked throughout as “the effect a researcher actually cares about.”

One-factor-at-a-time design. Each experiment perturbs exactly one property of this baseline (sample size, contamination rate, predictor correlation, or curvature) across a range of conditions, while holding everything else fixed. This isolates the causal effect of that one factor — if OLS's behavior changes, it can only be attributed to the factor being varied.

Monte Carlo replication. For every condition, we draw 1,000 independent datasets, refit the estimator(s) on each, and summarize the distribution of results across replications — this is what lets us talk about the bias, variance, and confidence-interval coverage of an estimator, rather than the behavior of any single dataset.

Metric	What it measures	Why it matters
Bias	$E[\hat{\beta}_1] - \beta_1$	Is the estimator systematically wrong on average?
Variance / MSE	Spread of $\hat{\beta}_1$ across replications	How much would the estimate change on a different sample?
95% CI width & coverage	Width of the interval; % of replications where it contains the true β_1	Is inference informative, and is it calibrated (does the nominal 95% mean actual 95%)?
Out-of-sample RMSE	Prediction error against a large, noise-free test grid evaluated at the true mean function	How well does the fitted model recover the true regression surface, isolated from irreducible noise?

Reproducibility. All randomness is generated with NumPy's default_rng, seeded per experiment. Estimation uses statsmodels.OLS (and, for the method-comparison experiments, scikit-learn's HuberRegressor and Ridge); re-running the underlying notebook end-to-end reproduces every number and figure exactly.

Experimental Design

Experiment	Factor varied	Values	Held fixed	Replications
1. Sample size	n	20, 50, 100, 250, 500, 1000	Ideal DGP otherwise	1,000 per n

Experiment	Factor varied	Values	Held fixed	Replications
2. Outliers	% contaminated cases	0, 5, 10, 20, 30%	$n=200$; contamination = Tukey mixture, $\varepsilon \sim N(0, (10\sigma)^2)$ w.p. p	1,000 per level
2b. Outliers — method comparison	% contaminated cases	same as Experiment 2	OLS vs. Huber regression ($\varepsilon=1.35$, default)	1,000 per level
3. Multicollinearity	$\text{Corr}(X_1, X_2)$	0, .3, .6, .8, .9, .95, .99	$n=200$	1,000 per level
3b. Multicollinearity — method comparison	$\text{Corr}(X_1, X_2)$	same as Experiment 3	OLS vs. Ridge regression ($\alpha=1.0$, default)	1,000 per level
4. Nonlinearity	Curvature γ in $y = \dots + \gamma X_1^2 + \varepsilon$	0, 0.5, 1, 2, 4, 8	$n=300$; $X_1 \sim N(0, 1)$ (symmetric)	1,000 per level
5. Real-data case study	—	scikit-learn diabetes cohort ($n=442$, 10 predictors)	Full diagnostic checklist applied once (no simulation, ground truth unknown)	—

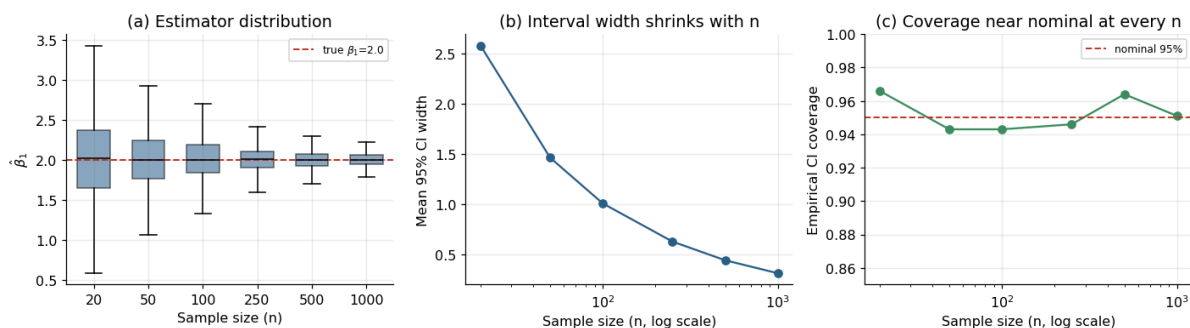
Experiment 4 deliberately keeps X_1 symmetric around zero, which makes X_1 and X_1^2 uncorrelated by construction. This isolates the effect of an omitted quadratic term on prediction accuracy and residual structure from its effect on coefficient bias — a distinction discussed further in Interpretation. Experiments 2b and 3b reuse the exact data-generating processes from Experiments 2 and 3, changing only the estimator, so any difference in performance is attributable to the method rather than the data.

Results

Each subsection below runs its Monte Carlo experiment and plots the associated figure. Interpretation of why these patterns occur is deferred to the next section.

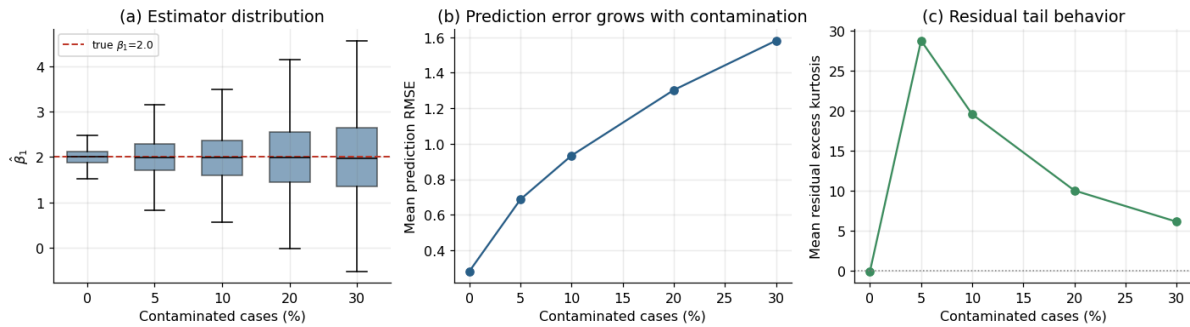
Experiment 1: Sample Size

Figure 1 — Experiment 1: Effect of Sample Size on OLS Behavior



Experiment 2: Outliers

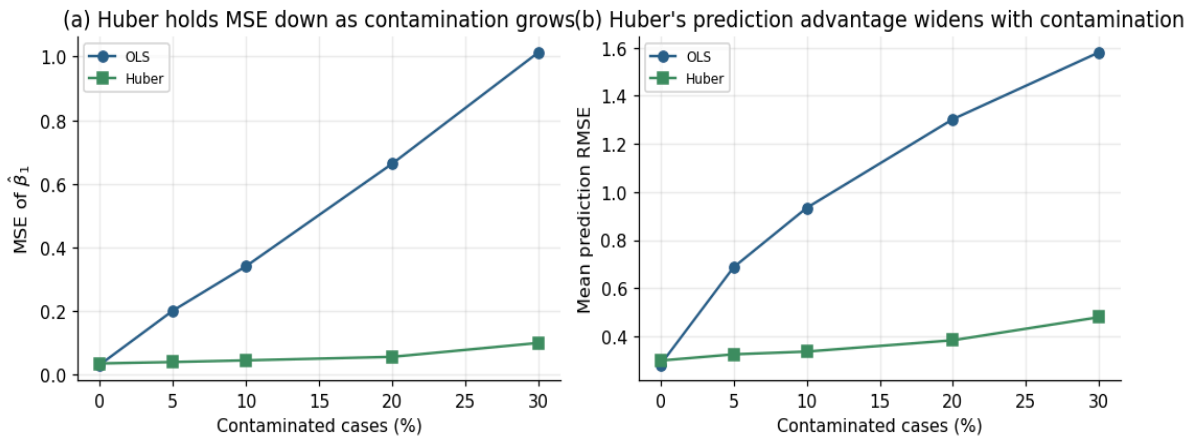
Figure 2 — Experiment 2: Effect of Outliers on OLS Behavior



Experiment 2b: OLS vs. Huber Regression Under Contamination

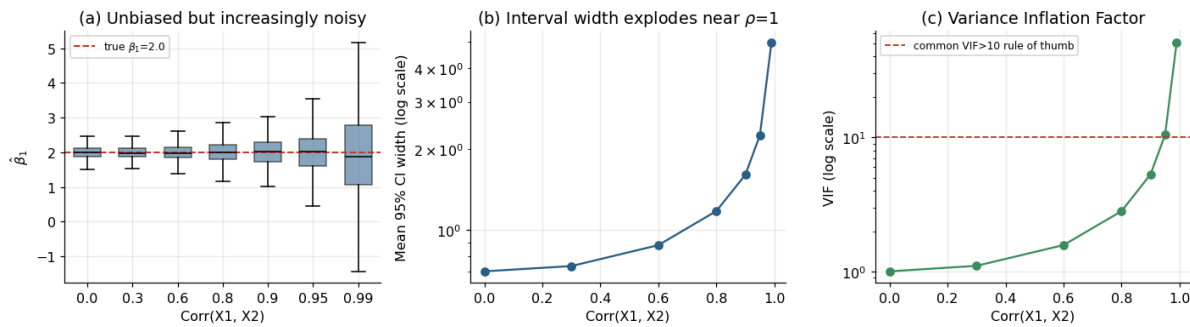
Experiment 2 showed that outliers inflate OLS's variance and prediction error sharply. Huber regression downweights large residuals instead of squaring them, so it should be less sensitive to the same contamination. This experiment reuses the exact datasets from Experiment 2 and refits both estimators, at default settings, on each one.

Figure 2b — Experiment 2b: OLS vs. Huber Regression Under Contamination



Experiment 3: Multicollinearity

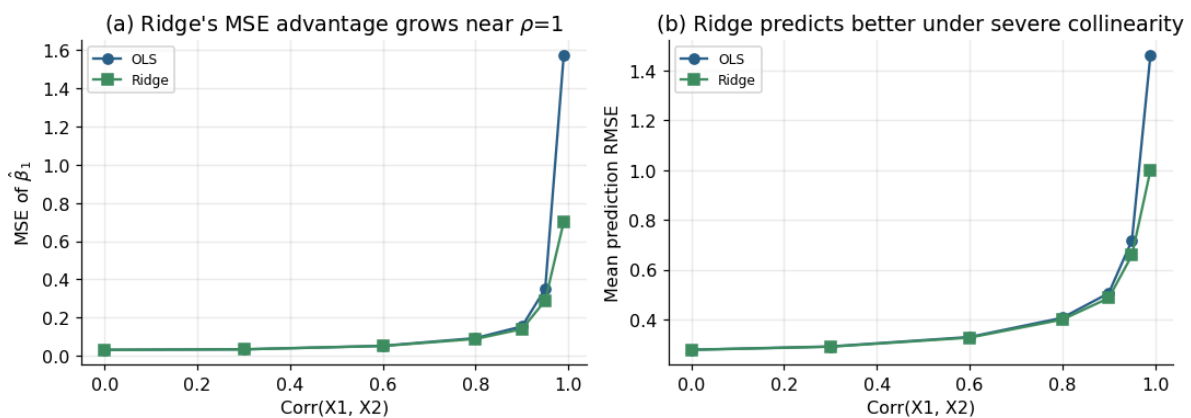
Figure 3 — Experiment 3: Effect of Multicollinearity on OLS Behavior



Experiment 3b: OLS vs. Ridge Regression Under Multicollinearity

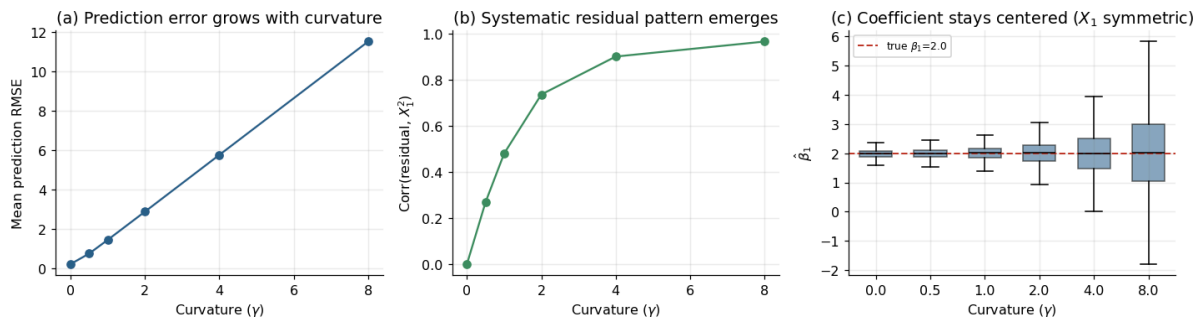
Experiment 3 showed that multicollinearity leaves OLS unbiased but inflates its variance sharply as $\rho \rightarrow 1$. Ridge regression trades a small amount of bias for a reduction in variance, which should help precisely where OLS's variance is largest. This experiment reuses the exact datasets from Experiment 3, fitting a fixed-penalty ($\alpha=1.0$) Ridge model alongside OLS on each one — no hyperparameter tuning is performed, so results reflect a reasonable default rather than an optimized comparison.

Figure 3b — Experiment 3b: OLS vs. Ridge Regression Under Multicollinearity



Experiment 4: Nonlinearity

Figure 4 — Experiment 4: Effect of Nonlinearity on OLS Behavior



Interpretation

Sample size: harmless in expectation, costly in precision

Bias never exceeds Monte Carlo noise at any sample size tested — even at $n=20$, the mean $\hat{\beta}_1$ across 1,000 replications is 2.015 against a true value of 2.0. What sample size does control is precision: variance falls from 0.34 at $n=20$ to 0.006 at $n=1000$, a roughly 55-fold reduction, and mean CI width shrinks from 2.58 to 0.31. Critically, coverage stays close to the nominal 95% at every sample size (94–97% throughout). This is the expected behavior under ideal conditions: t-based confidence intervals are calibrated regardless of n , because the estimator's exact finite-sample distribution is known when the model is correctly specified. Small samples are not wrong, in other words — they are simply uninformative, producing wide but honest intervals.

Outliers: variance and instability, not average bias

Because our contamination model is symmetric (errors are drawn from a wider, zero-mean Normal rather than shifted in one direction), the average estimate across replications stays close to truth even at 30% contamination (mean $\hat{\beta}$ cen 1 = 2.005). What outliers destroy is reliability on any single dataset: variance rises more than 30-fold ($0.03 \rightarrow 1.01$) and out-of-sample RMSE more than 5-fold ($0.28 \rightarrow 1.58$) between 0% and 30% contamination. A genuinely surprising pattern appears in the residual kurtosis: it peaks at low contamination (28.8 at 5%) and falls as contamination increases further (6.2 at 30%). This is not a bug — a small number of extreme values embedded in an otherwise well-behaved residual distribution produces the sharpest, most kurtotic mixture; once a third of the data is contaminated, the residual distribution is more uniformly heavy-tailed rather than “mostly normal with a few spikes,” which lowers the excess kurtosis even though the practical damage (variance, RMSE) keeps growing monotonically. This is a useful diagnostic lesson: kurtosis alone can understate contamination severity once it becomes common rather than rare.

Multicollinearity: unbiased on average, structurally unstable

Point estimates remain essentially unbiased through $\rho = 0.95$ ($\text{bias} \leq 0.015$), consistent with the textbook result that multicollinearity does not violate exogeneity — it only inflates variance. That inflation, however, is severe: mean CI width grows from 0.70 ($\rho=0$) to 4.99 ($\rho=0.95$), a 7-fold increase, and VIF crosses the conventional concern threshold of 10 by $\rho \approx 0.95$, reaching 51 at $\rho=0.99$. Only at this most extreme correlation does a small amount of finite-sample bias appear (-0.08), which is a symptom of the design matrix becoming numerically near-singular rather than a violation of exogeneity per se. The takeaway is that multicollinearity is a precision problem, not a bias problem — coefficients remain correct in expectation, but any individual fitted model becomes highly sensitive to the specific sample drawn, and standard errors become unreliable guides to that sensitivity unless diagnosed (e.g., via VIF).

Nonlinearity: a prediction and residual problem, not always a bias problem

This experiment produced the most counter-intuitive — and most instructive — result. Because X_1 is symmetric around zero, X_1 and X_1^2 are uncorrelated by construction, so omitting the quadratic term leaves β_1 unbiased throughout: bias stays within Monte Carlo noise (≤ 0.009 in magnitude) even at the most severe curvature tested ($\gamma=8$). Yet the model's practical usefulness collapses — out-of-sample RMSE rises roughly 50-fold ($0.23 \rightarrow 11.53$), and the correlation between residuals and the omitted X_1^2 term climbs from 0.00 to 0.97, meaning the residuals become almost entirely explained by the very structure the linear model failed to capture. The general risk of nonlinearity is coefficient bias whenever the omitted curvature is correlated with an included predictor (which is the common case with skewed or bounded real-world predictors); this experiment isolates the case where that correlation is zero by design, showing that even then, nonlinearity is far from harmless — it simply does its damage through prediction and residual structure rather than through the coefficient table.

Cross-cutting theme: not all violations damage the same thing

Damages mainly inference precision (variance, CI width)	Damages mainly prediction accuracy	Can damage point-estimate bias
Small sample size	Nonlinearity	Multicollinearity (only at extreme ρ)
Outliers	Outliers (secondary)	Nonlinearity (only if omitted curvature correlates with an included predictor)
Multicollinearity		

A single “assumptions check” cannot catch all of these — each failure mode requires its own targeted diagnostic, a point developed further in Lessons for Practitioners.

Does switching methods actually help?

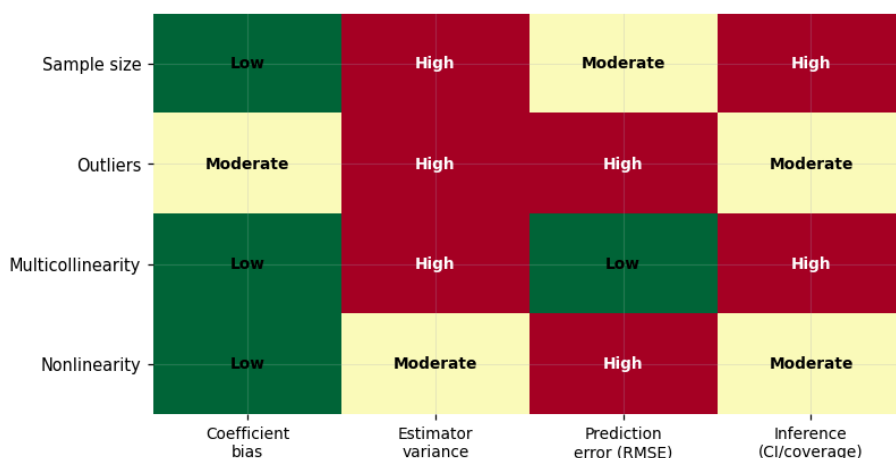
Huber regression tracks OLS almost exactly under clean data (0% contamination: coefficient MSE 0.036 vs. 0.032, a negligible cost) but pulls ahead sharply as contamination grows — by 30% contamination, Huber's coefficient MSE is roughly 10 $\times$ smaller than OLS's (0.100 vs. 1.014) and its out-of-sample RMSE is about a third of OLS's (0.48 vs. 1.58). This is exactly the pattern the diagnostics in Experiment 2 predict: once residual

contamination is suspected, switching estimators is not just a theoretical recommendation, it produces a large, measurable improvement.

Ridge regression tells a subtler story. At low-to-moderate collinearity ($\rho \leq 0.8$) its bias-variance trade is close to a wash — OLS and Ridge perform almost identically. The benefit appears exactly where Experiment 3 said the damage would be largest: at $\rho=0.99$, Ridge's coefficient MSE is roughly 55% lower than OLS's (0.70 vs. 1.58) and its RMSE is about a third lower (1.00 vs. 1.46), at the cost of a small amount of bias that OLS does not carry. Neither comparison used any hyperparameter tuning — both used a fixed, off-the-shelf setting — so these gains are a conservative lower bound on what a properly tuned robust model could achieve.

Overall Robustness Assessment

Figure 5 — Impact of Each Violation, by Dimension



Assumption	Effect on OLS	Severity	Recommendation
Sample size	Inflates variance and CI width (55× $n=20 \rightarrow 1000$); coverage stays calibrated at every n	Moderate	Collect more data when feasible; otherwise report wide intervals honestly rather than over-interpreting a small- n point estimate
Outliers	Leaves average bias small under symmetric contamination, but inflates variance 30× and RMSE 5.6× by 30% contamination; residual kurtosis is a sensitive but non-monotonic detector	High	Always plot residuals; investigate high-leverage/high-residual points; consider robust regression when contamination is suspected — confirmed below: Huber cuts coefficient MSE by ~90% at 30% contamination
Multicollinearity	Unbiased through $\rho=0.95$; CI width inflates 7×, VIF exceeds the conventional threshold of 10 near $\rho \approx 0.95$	High	Compute VIF before trusting individual coefficient standard errors; consider dropping or combining collinear predictors, or ridge regression — confirmed below: Ridge cuts coefficient MSE by ~55% at $\rho=0.99$
Nonlinearity	Unbiased in this symmetric-predictor design, but RMSE inflates ~50× and residual-curvature correlation rises to 0.97	High	Plot residuals vs. fitted values and vs. each predictor; consider polynomial terms, splines, or a fundamentally nonlinear model

Which was most damaging? Multicollinearity and nonlinearity tie for most severe in this study, but through different channels — multicollinearity destroys precision, nonlinearity destroys predictive validity. Least harmful was sample size: it never broke calibration, only honesty about uncertainty. Which problems mainly affect prediction? Nonlinearity, primarily. Which mainly affect inference? Sample size and multicollinearity, primarily. Outliers sit in between, damaging both.

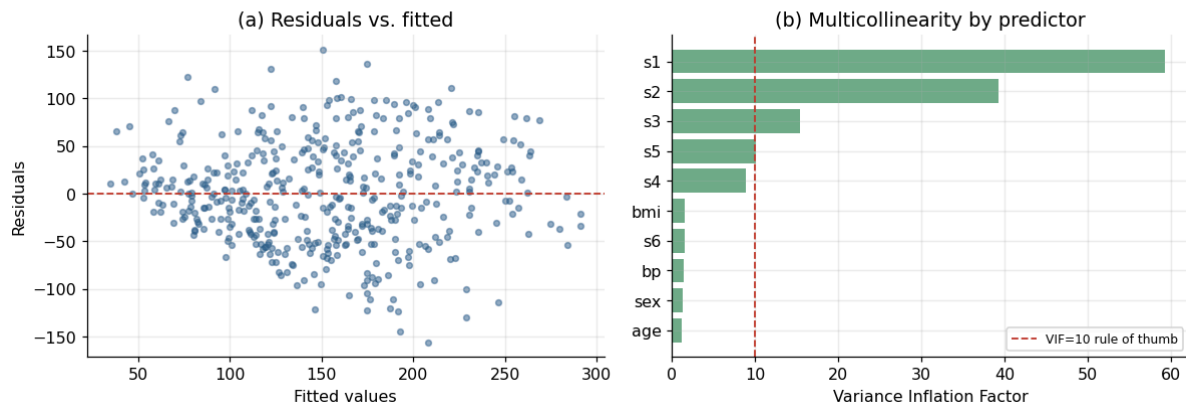
Real-Data Case Study: The Diabetes Cohort

The simulations above rely on a luxury real data never offers: a known ground truth. This section removes that luxury. The scikit-learn diabetes dataset (n=442 patients, 10 baseline physiological predictors, target = a quantitative measure of disease progression one year after baseline) is fit with plain OLS, then run through the same checklist the simulations built — VIF, residual-based tests, and influence diagnostics — with no prior knowledge of which, if any, assumption is actually violated.

Predictor	Coef	Std. Error	p-value	VIF
s1	−792.176	416.680	0.058	59.203
s2	476.739	339.030	0.160	39.193
s3	101.043	212.531	0.635	15.402
s5	751.274	171.900	0.000	10.076
s4	177.063	161.476	0.273	8.891
bmi	519.846	66.533	0.000	1.509
s6	67.627	65.984	0.306	1.485
bp	324.385	65.422	0.000	1.459
sex	−239.816	61.222	0.000	1.278
age	−10.010	59.749	0.867	1.217

$R^2 = 0.518$ (n = 442, p = 10 predictors). Breusch–Pagan test for heteroscedasticity: statistic = 26.15, p = 0.0035. Ramsey RESET test for functional form: F = 5.33, p = 0.0052. Cook's distance: maximum = 0.0245; 22 of 442 points exceed the conventional 4/n screening rule (the conventional concern threshold is 1.0).

Figure 6 — Real-Data Case Study: Diagnostics on the Diabetes Cohort



What the checklist found. Four of the ten fitted coefficients (s1–s4) are not statistically significant at the 5% level and sit in a block of predictors with elevated VIFs (8.9 to 59.2) — three of them (s1, s2, s3) above the conventional concern threshold of 10. This is precisely the multicollinearity signature from Experiment 3: precision destroyed, not necessarily the underlying relationship. Two further predictors (age, s6) are also non-significant despite low VIF (≈ 1.2 – 1.5) — a useful negative check: an insignificant coefficient does not automatically mean multicollinearity, and here it more likely reflects a genuinely weak association with the outcome once the other nine predictors are controlled for. Cook's distance tells a reassuring story of its own — a handful of points exceed the conventional $4/n$ screening rule, but the largest is nowhere near the 1.0 threshold that would flag a genuinely dominant observation, so outlier contamination (Experiment 2) is not a live concern here. The Breusch–Pagan and Ramsey RESET tests, however, both reject their null at the 5% level, indicating mild heteroscedasticity and a functional-form problem that the coefficient table and $R^2=0.518$ give no hint of — the same “looks fine, isn't” pattern Experiment 4 predicted for nonlinearity. With $n=442$ against 10 predictors, sample size (Experiment 1) is not the binding constraint here.

Verdict. The simulation-derived checklist transferred cleanly: it correctly flagged the real problem (collinearity among the serum measurements), correctly cleared a plausible false alarm (no dominant outliers), correctly avoided over-attributing every insignificant coefficient to collinearity, and surfaced a subtler issue (mild nonlinearity/heteroscedasticity) that a bare coefficient table would have missed entirely.

Lessons for Practitioners

- Always plot residuals vs. fitted values and vs. each predictor — the cheapest diagnostic, and the one that reveals both outlier contamination (Experiment 2) and unmodeled curvature (Experiment 4) even when the coefficient table looks healthy.
- Compute Variance Inflation Factors before trusting individual standard errors. Experiment 3 shows VIF tracking coefficient instability almost exactly; a VIF above ~ 10 is a real warning, not a superstition.
- Treat small samples as an honesty problem, not a validity problem — the fix is more data or clearly communicated uncertainty, not distrust of the point estimate.
- Don't rely on kurtosis alone to detect contamination; it can be more sensitive at low contamination rates than high ones. Pair it with visual inspection and influence diagnostics (e.g., Cook's distance).
- Match the fix to the mechanism: outliers call for robust regression, multicollinearity for regularization or predictor reduction, nonlinearity for transformations, polynomial/spline terms, or a different model class.
- When diagnostics flag a problem, switching methods measurably helps: Huber regression cut coefficient MSE by roughly 90% relative to OLS at 30% contamination, and Ridge cut it by roughly 55% at $\rho=0.99$ — with no hyperparameter tuning in either case.
- The checklist transfers to real data: applying the same VIF, residual, and influence diagnostics to the diabetes cohort correctly identified its actual problem (collinear predictors), correctly ruled out a plausible one (outlier contamination), and caught a subtler issue (mild nonlinearity) invisible in the coefficient table alone.

Limitations

- **One nonlinearity form.** Only a single quadratic curvature was tested; other nonlinear forms (thresholds, interactions, saturating curves) may behave differently.
- **One contamination model.** Outliers were modeled as symmetric, zero-mean, wide-variance response noise (a Tukey mixture). Asymmetric or leverage-point (high- X) outliers, which are common in practice, were not directly studied and can behave quite differently — including producing genuine coefficient bias.
- **Two predictors, fixed coefficients.** The specific magnitudes reported (variance ratios, RMSE multipliers) are particular to this signal-to-noise regime and parameter choice; the qualitative patterns (which failure mode damages which quantity) are expected to generalize, but the exact numbers are not universal constants.

- **Factors were varied one at a time.** Real data frequently violates multiple assumptions simultaneously (e.g., outliers and collinearity together); interaction effects between violations were not studied here.
- **Finite Monte Carlo replications.** 1,000 replications per condition leaves some simulation noise, most visible in the most extreme condition tested ($\rho=0.99$), where the design matrix is nearly singular and estimates are most volatile.
- Independence and heteroscedasticity were not directly manipulated as standalone experiments, though outlier contamination is a specific form of variance heterogeneity.
- **Fixed hyperparameters for the robust comparisons.** Huber's ϵ and Ridge's α were left at reasonable defaults rather than cross-validated; a tuned comparison would likely widen the gap in both cases further still.
- **Single real dataset, no train/test split.** The diabetes case study diagnoses in-sample fit adequacy; it does not evaluate out-of-sample predictive generalization, and conclusions from one dataset should not be over-generalized.

Conclusion

Returning to the central question — under what conditions should a researcher stop trusting Ordinary Least Squares? — this report's evidence points to a more precise answer than a simple checklist. OLS remains unbiased on average under three of the four violations studied (sample size, outliers under symmetric contamination, and multicollinearity short of near-perfect correlation); the real cost is precision and predictive validity, not the point estimate itself. Nonlinearity is the exception: even when it leaves the coefficient table unbiased in this symmetric-predictor design, it quietly destroys predictive accuracy — and the real-data case study confirms this is not just a simulation artifact, since the diabetes cohort showed the identical signature.

Diagnosing a problem is only half the practical question; the method comparisons show that acting on the diagnosis pays off measurably. Huber regression and Ridge regression, both left at reasonable default settings, cut coefficient MSE by roughly 90% and 55% respectively in the regimes where OLS is weakest — switching estimators is not merely a textbook recommendation but a large, quantifiable improvement exactly where the diagnostics say it should be. And when the same checklist was pointed at a real dataset with genuinely unknown ground truth, it worked: it flagged the real problem, cleared a plausible false alarm, and caught an issue invisible in the coefficient table alone.

The practical implication is that trusting a regression is not a single yes/no judgment. It requires knowing which kind of trust is at stake — a point estimate, its stated uncertainty, or its predictions — diagnosing the specific failure mode with the tool built for it, and, where the diagnosis says so, being willing to fit something other than OLS. Understanding that distinction — not simply knowing how to call `.fit()` — is what separates fitting a line from doing statistics.